

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T36 · Graphing Inequalities

Questions

Student book · questions with writing space

■ 14 pages

FAMILY MAP

Graphing Inequalities

F01

Recover inequalities from a shaded region

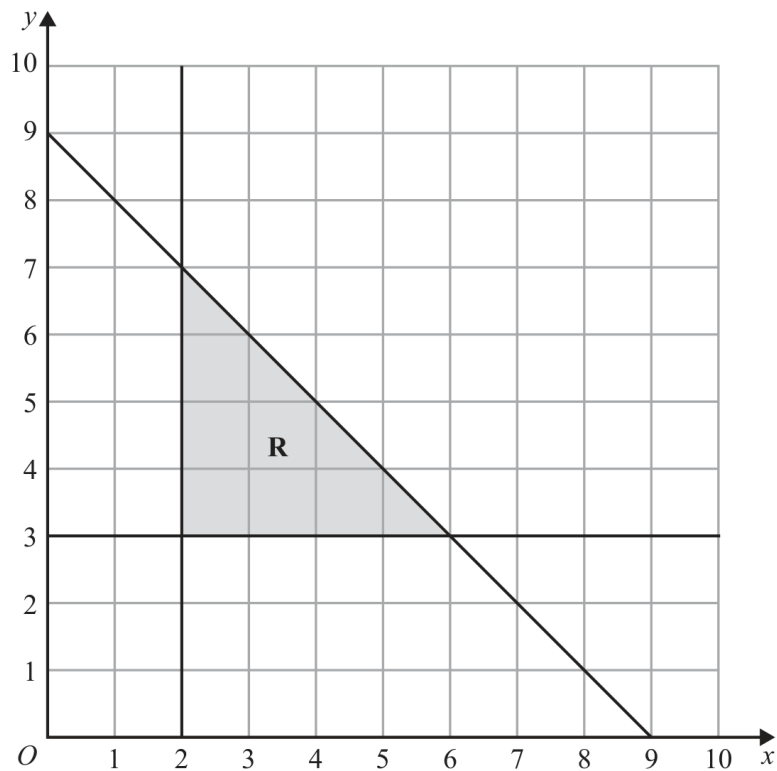
5 Q | 16 marks

F02

Construct the feasible region from inequalities

8 Q | 26 marks

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



(b) Write down three inequalities that define the region $\mathbf{R}$

Question 13

4MA1-2H Q2

MODULAR UNIT 2

4 marks

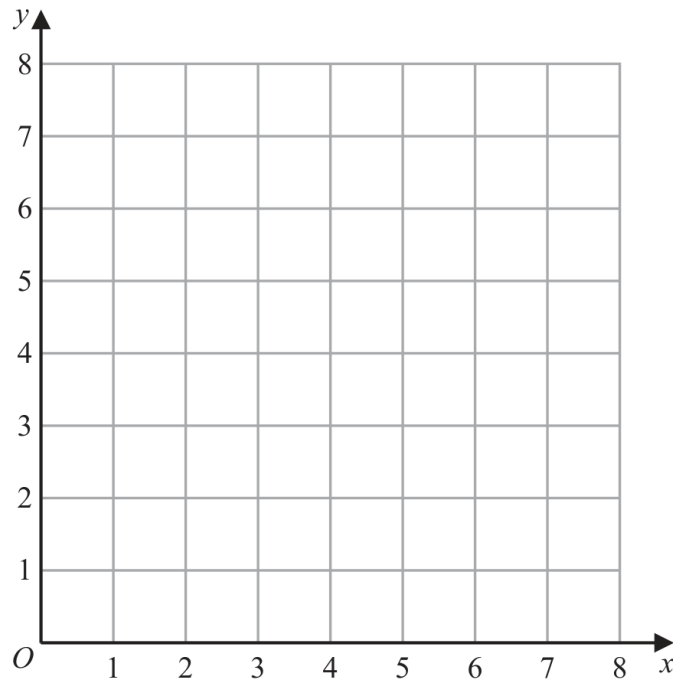
2 (a) On the grid, draw the straight line with equation

(i) $y = 2$

(ii) $x = 6$

(iii) $y = x + 1$

Label each line with its equation.



(3)

(b) Show, by shading on the grid, the region that satisfies all three of the inequalities

$y \geq 2$

$x \leq 6$

$y \leq x + 1$

Label the region **R**

(1)

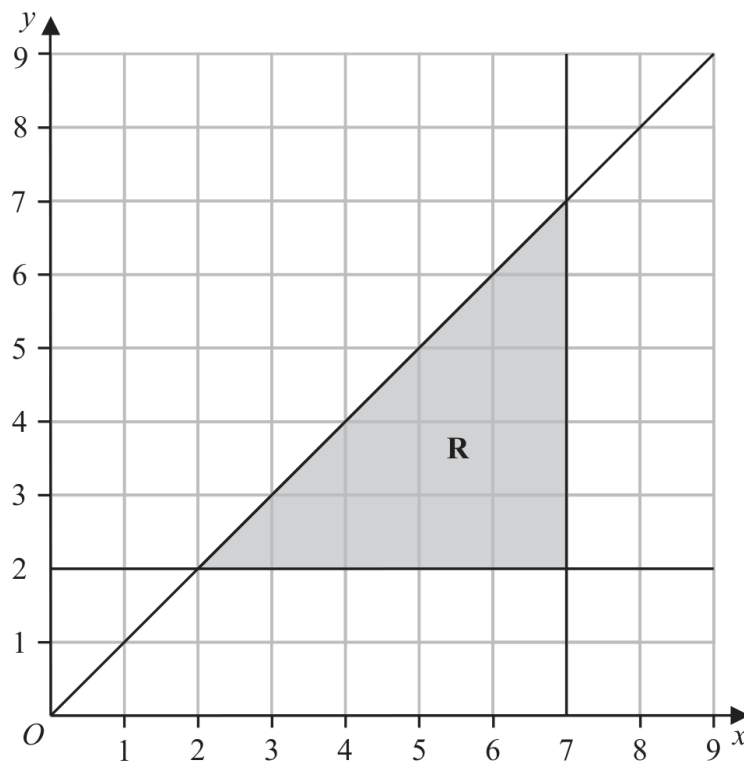
Question 05

4MA1-1H Q8 (b)

MODULAR UNIT 2

3 marks

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



(b) Write down the three inequalities that define the region **R**

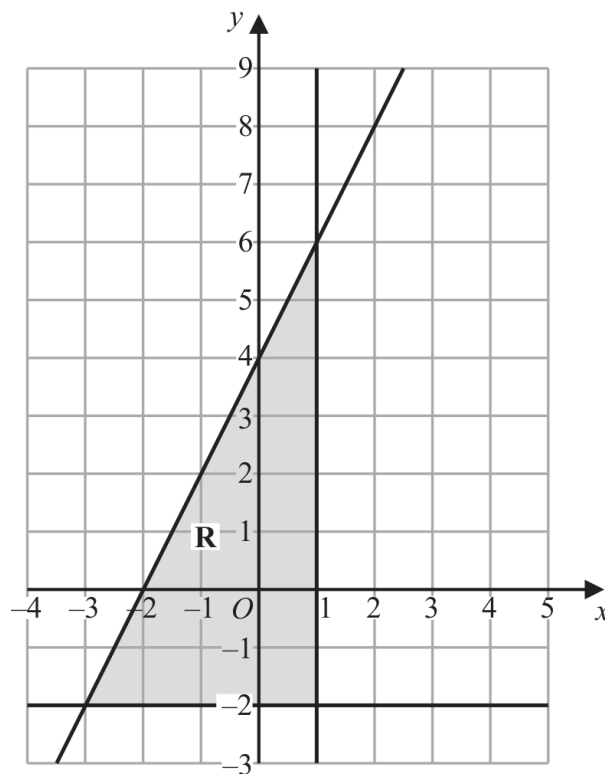
.....
.....
.....
(3)

Question 06

4MA1-1H Q7

MODULAR UNIT 2

4 marks



The region **R**, shown shaded in the diagram, is bounded by three straight lines.

Find the inequalities that define **R**

Question 21

4MA1-2H Q7 M01, M02

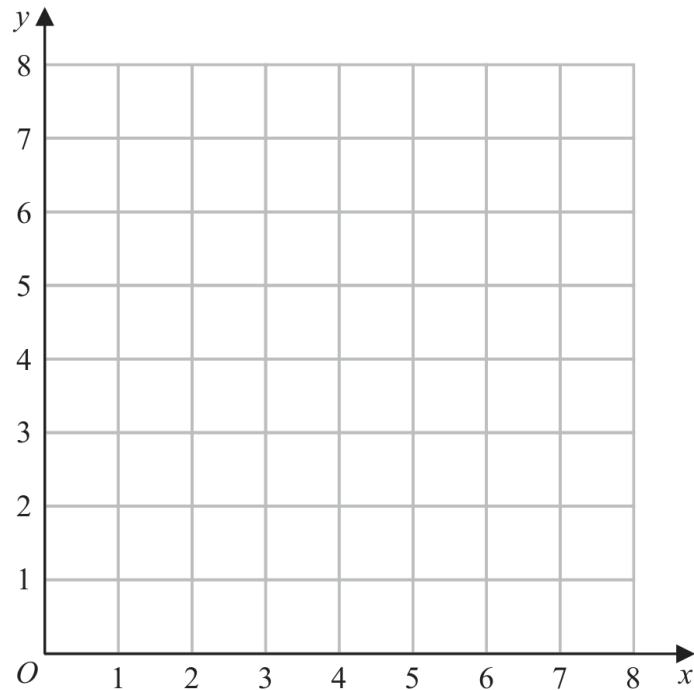
MODULAR UNIT 2

4 marks

- 7 (a) On the grid, draw the straight line with equation

(i) $x = 3$ (ii) $y = 1$ (iii) $x + y = 7$

Label each line with its equation.



(3)

- (b) Show, by shading on the grid, the region that satisfies all three of the inequalities

$$x \geq 3 \qquad y \geq 1 \qquad x + y \leq 7$$

Label the region **R**

Question 21

4MA1-2H Q2 Q2(a), Q2(b)

MODULAR UNIT 2

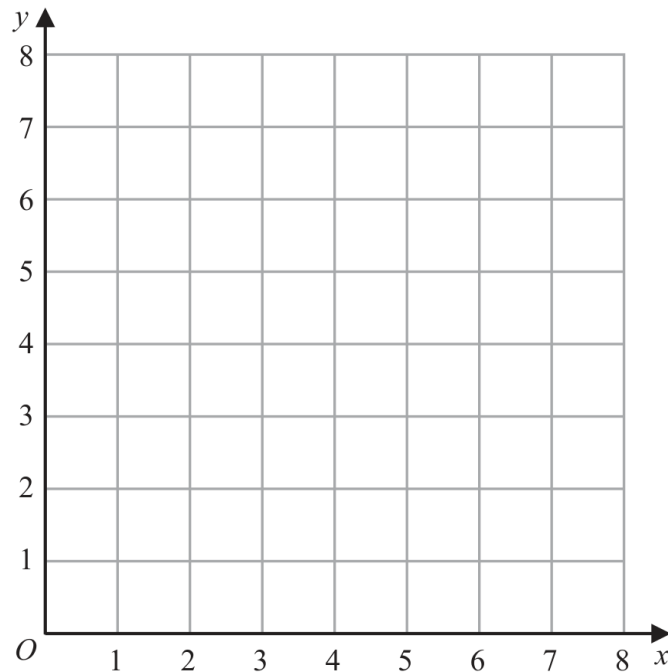
4 marks

- 2 (a) On the grid, draw and label with its equation the straight line with equation

(i) $y = 1$

(ii) $x = 2$

(iii) $x + y = 7$



(3)

- (b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$y \geq 1$

$x \geq 2$

$x + y \leq 7$

Label the region **R**.

(1)

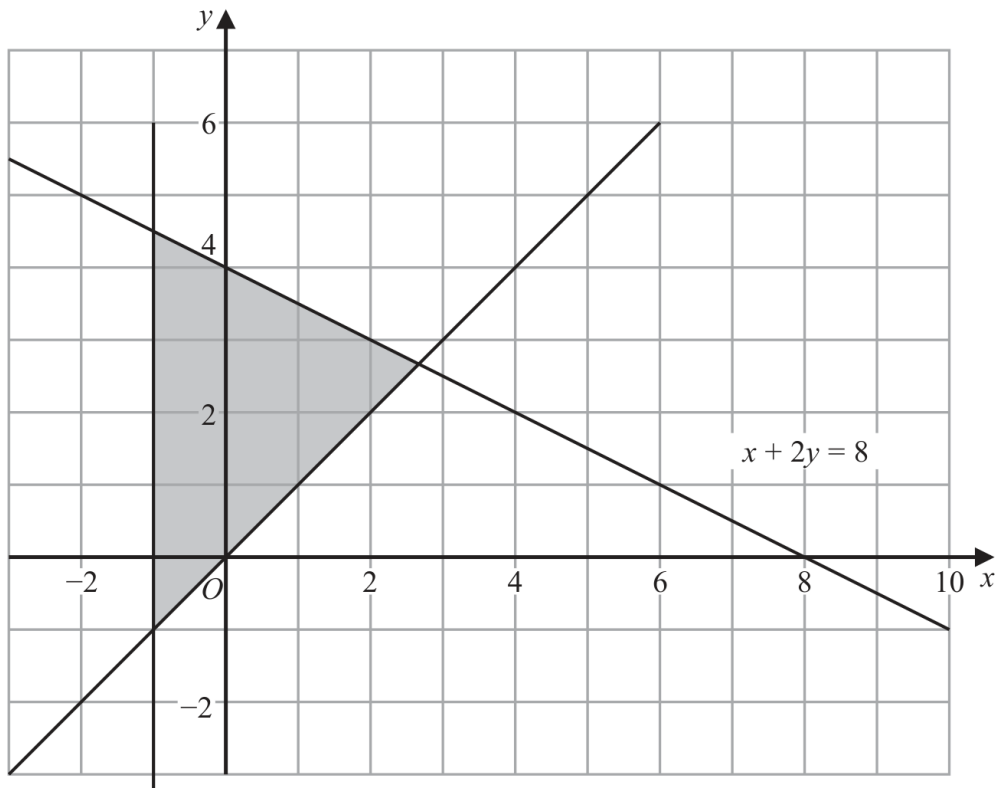
Question 27

4MA1-2H Q13

MODULAR UNIT 2

3 marks

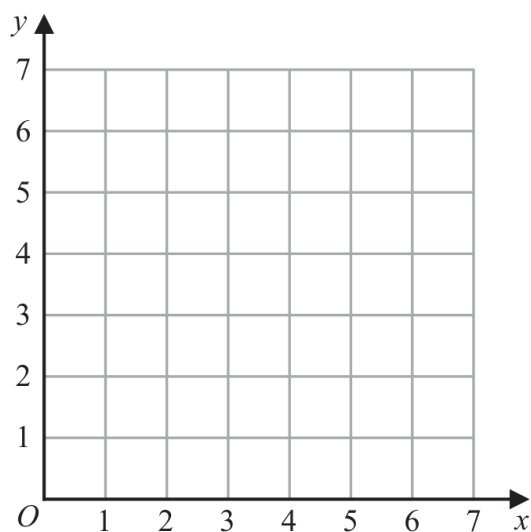
- 13 The shaded region in the diagram is bounded by three lines.
The equation of one of the lines is given.



Write down three inequalities that define the shaded region.

-
- A coordinate plane with x and y axes ranging from -6 to 6. The grid lines are spaced at intervals of 1 unit. The x-axis is labeled with -6, -4, -2, 2, 4, 6. The y-axis is labeled with -6, -4, -2, 2, 4, 6. The origin is labeled O. Two lines are graphed: a solid line with a negative slope passing through (0, 4) and (4, 0), and a solid line with a positive slope passing through (0, -2) and (6, 0). The region below the first line and above the second line is shaded gray. A label $y = \frac{1}{3}x - 2$ is placed near the second line.

4



(a) On the grid, draw and **label** the straight line with equation

(i) $x = 1.5$

(ii) $y = x$

(iii) $x + y = 6$

(3)

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

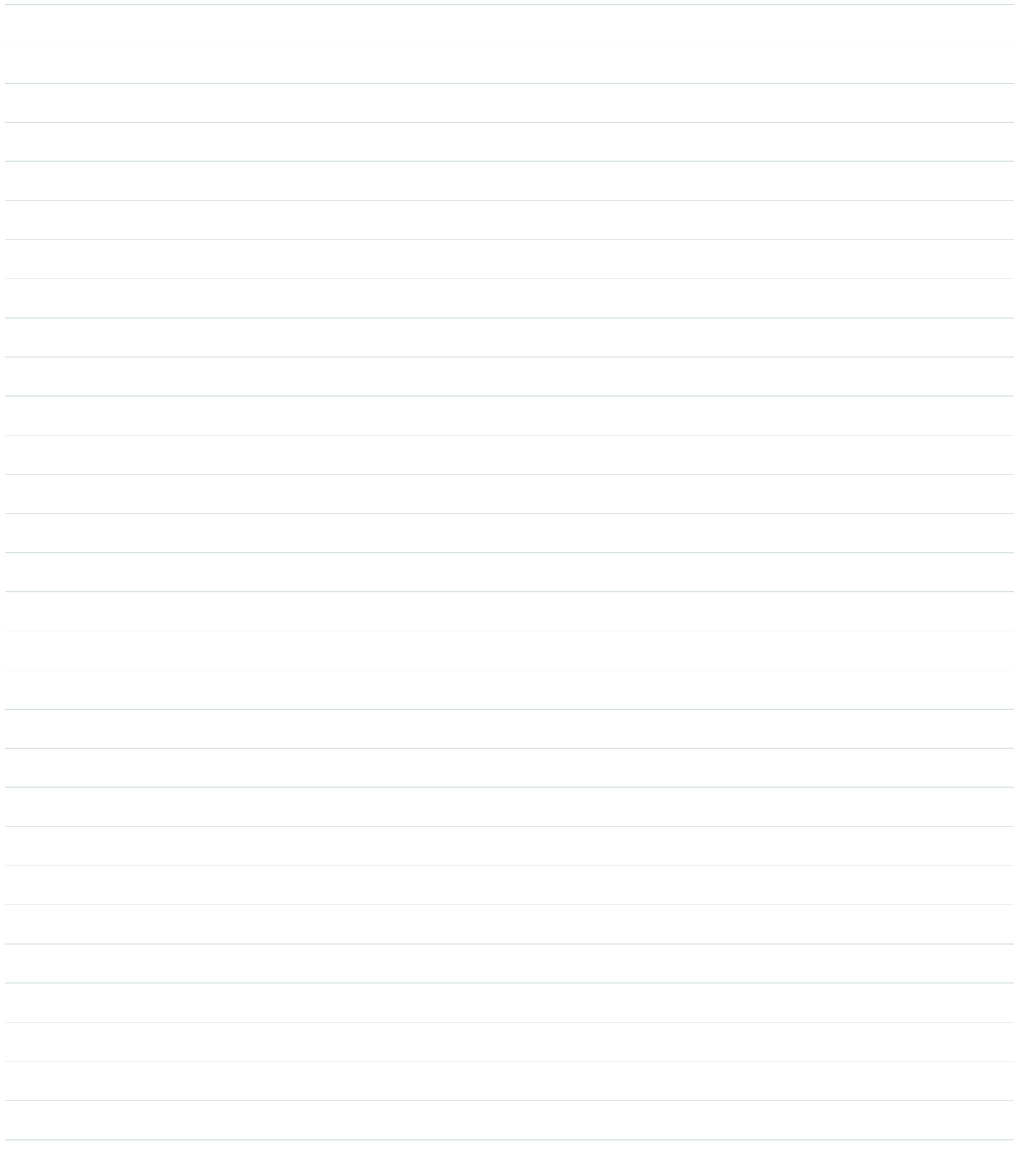
$$x \geq 1.5$$

$$y \geq x$$

$$x + y \leq 6$$

Label the region **R**.

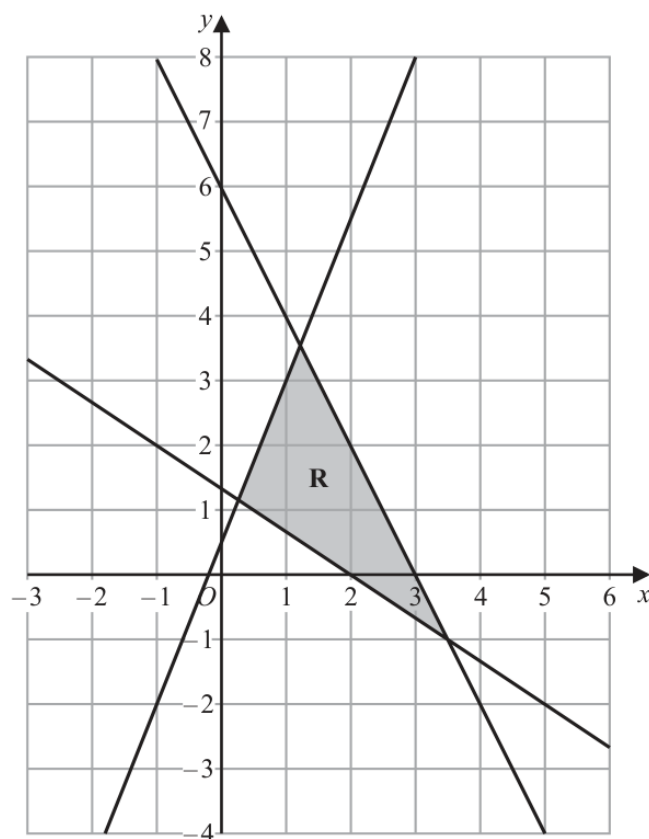
(1)



A Cartesian coordinate system with x and y axes. The x-axis is labeled from -4 to 4, and the y-axis is labeled from -4 to 4. A grid is shown with major lines every 1 unit and minor lines every 0.2 units. Two lines are plotted: a line with equation $y = 2x + 2$ and a line with equation $y = -x + 4$. The feasible region R is the shaded area bounded by these two lines and the x-axis. The region is a triangle with vertices at $(-2, 0)$, $(0, 2)$, and $(4, 0)$. The origin is labeled O .

Write down the three inequalities that define $\mathbf{R}$.

11



The region **R**, shown shaded in the diagram, is bounded by the straight lines with equations

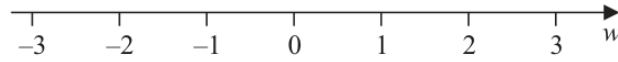
$$2x + y = 6$$

$$2y = 5x + 1$$

$$3y + 2x = 4$$

Write down the three inequalities that define **R**

(1)



The graph shows a Cartesian coordinate system with x and y axes ranging from -5 to 5. A shaded region R is defined by the following boundaries:

- The y-axis ($x = 0$).
- The horizontal line $y = 4$.
- The diagonal line $y = x$.
- The vertical line $x = -3$.

The region R is shaded in light gray. The label R is placed inside the shaded area in the second quadrant.

(b) Write down the three inequalities that define the region $\mathbf{R}$